

Part II: Endogenous Skill

Reading the equations as economic stories

An intuition-first lecture on Section 4 of the tutorial

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Human–AI Productivity Paradoxes

The paradox comes from the weights, not the groups

At every fixed skill state, AI weakly raises optimized productivity:

$$\frac{\partial p^*(s_k, a)}{\partial a} \geq 0.$$

Yet long-run productivity is a weighted average,

$$\mathcal{P}(a) = \sum_{k=1}^N \pi_k(a) p^*(s_k, a),$$

and the weights $\pi_k(a)$ also change with AI.

Bottom line

AI can improve productivity within every skill state while shifting enough probability toward low skill to reduce the aggregate.

Three questions organize the entire lecture

1. Why does more AI reduce the incentive to learn?
2. How does lower effort become a worse long-run skill distribution?
3. When can that composition loss dominate AI's direct productivity gain?

The answer in one line

substitution $\longrightarrow$ deskilling $\longrightarrow$ composition reversal

Recall the static result: AI first replaces effort

At skill s_k , optimal effort is

$$e_k(a) = e^*(s_k, a) = (x^* - s_k - a)_+.$$

As long as effort is positive,

$$\frac{de_k(a)}{da} = -1.$$

Economic meaning

One additional unit of AI replaces exactly one unit of human effort until effort reaches zero. Total productive input remains at x^* during this replacement phase.

At a fixed skill, AI never lowers output

The optimized state-level productivity is

$$p_k^*(a) = \max\{p(x^*), p(s_k + a)\}.$$

Therefore,

$$\frac{dp_k^*(a)}{da} \geq 0.$$

Economic meaning

The paradox cannot be a failure of the static optimizer. It must come from a change in how often the agent occupies each skill state.

Effort is both a current cost and a learning input

The dynamic model adds one link:

$$e_k(a) \mapsto \lambda(e_k(a)).$$

Here $\lambda(e)$ is increasing and concave:

$$\lambda'(e) \geq 0, \quad \lambda''(e) \leq 0.$$

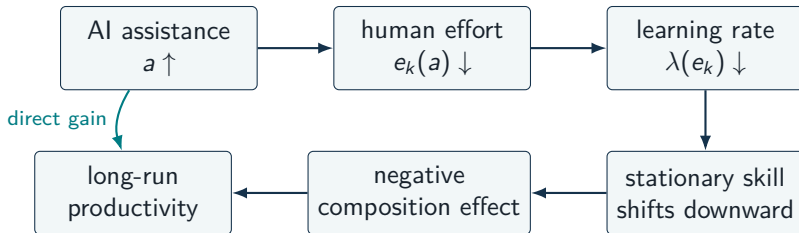
Economic meaning

More effort makes upward skill movement arrive faster, but each extra unit of effort adds less learning speed than the previous one.

Bottom line

AI crowds out the very activity that produces future skill.

The mechanism is a feedback loop



Bottom line

The theorem compares the positive direct path with the negative learning path.

From effort to long-run skill

Rates tell us how quickly the agent moves; stationary probabilities tell us where the agent spends time.

A transition rate is a speed, not a probability

If the agent is currently at s_i , then over a very short interval h ,

$$\Pr\{S(t+h) = s_j \mid S(t) = s_i\} = r_{ij}h + o(h).$$

Economic meaning

r_{ij} is the arrival speed of the move from i to j . The probability over a short interval is approximately rate $\times$ interval length.

Why can a rate exceed one?

A rate is measured per unit of time. A rate of 3 per year does not mean a probability of 3; over one week, the probability is approximately $3/52$.

The model has an upward clock and a downward clock

At skill state s_k ,

$$s_k \rightarrow s_{k+1} \text{ at rate } \lambda(e_k(a)), \quad s_k \rightarrow s_{k-1} \text{ at rate } \mu.$$

Economic meaning

Effort controls the speed of skill improvement. Skill depreciation arrives at the constant speed $\mu > 0$.

Bottom line

AI affects upward movement through effort, but it does not change the downward clock directly.

Competing clocks turn rates into intuitive probabilities

At an interior state, write $\lambda_k = \lambda(e_k(a))$. Then

$$\Pr(\text{next move is up}) = \frac{\lambda_k}{\lambda_k + \mu}, \quad \Pr(\text{next move is down}) = \frac{\mu}{\lambda_k + \mu}.$$

The expected waiting time until either move is

$$\frac{1}{\lambda_k + \mu}.$$

Economic meaning

A larger upward rate makes the next change more likely to be an improvement and makes that change arrive sooner.

Skill moves along a birth–death ladder



Economic meaning

Only neighboring skill moves are possible. Reaching a distant high-skill state requires a sequence of successful upward transitions.

Stationary probability is a long-run time share

Fix the AI level a . Define

$$\pi_k(a) = \text{long-run fraction of time spent at skill } s_k.$$

The probabilities satisfy

$$\pi_k(a) \geq 0, \quad \sum_{k=1}^N \pi_k(a) = 1.$$

Economic meaning

Stationarity does not mean the agent stops moving. It means that probability inflow and outflow balance, so the distribution no longer changes over time.

Detailed balance is an accounting identity for flows

Across the edge between s_k and s_{k+1} ,

$$\pi_k(a)\lambda(e_k(a)) = \pi_{k+1}(a)\mu.$$

Left-hand side

mass at $s_k \times$ upward speed
= upward flow per unit time.

Right-hand side

mass at $s_{k+1} \times$ downward speed
= downward flow per unit time.

Bottom line

In the long run, traffic across each edge must be equal in both directions.

The adjacent probability ratio summarizes one edge

Divide detailed balance by $\pi_k(a)\mu > 0$:

$$\frac{\pi_{k+1}(a)}{\pi_k(a)} = \frac{\lambda(e_k(a))}{\mu}.$$

Economic meaning

The relative mass on the higher state rises when learning is faster and falls when depreciation is faster.

Reading the ratio

If $\lambda(e_k)/\mu = 2$, state s_{k+1} receives twice as much stationary mass as state s_k . If the ratio is $1/2$, it receives half as much.

The product formula accumulates the odds of every climb

Repeatedly applying the adjacent ratio gives

$$\pi_k(a) = \pi_1(a) \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu}.$$

Economic meaning

To reach state s_k from the bottom, the process must cross every earlier edge. Each factor records how favorable one of those climbs is relative to decay.

Bottom line

A reduction in an early learning rate affects the stationary weight of every state above that edge.

Normalization turns positive weights into probabilities

The product formula determines probabilities only up to the common factor $\pi_1(a)$. The condition $\sum_k \pi_k = 1$ gives

$$\pi_1(a) = \left[1 + \sum_{k=2}^N \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu} \right]^{-1}.$$

Economic meaning

The expression in brackets is the total unnormalized mass. Its reciprocal rescales all weights so that they sum to one. The exponent -1 simply means “take the reciprocal.”

A two-state example makes the stationary logic concrete

Suppose

$$\lambda_1 = 0.30, \quad \mu = 0.10.$$

Then detailed balance implies

$$0.30\pi_1 = 0.10\pi_2 \implies \pi_2 = 3\pi_1.$$

Together with $\pi_1 + \pi_2 = 1$,

$$(\pi_1, \pi_2) = (0.25, 0.75).$$

Economic meaning

An upward clock three times faster than the downward clock places three times as much long-run mass on the high-skill state.

Long-run outcomes are stationary weighted averages

At state s_k , effort is $e_k(a)$ and productivity is $p_k^*(a)$. Therefore

$$\mathcal{E}(a) = \sum_{k=1}^N \pi_k(a) e_k(a), \quad \mathcal{P}(a) = \sum_{k=1}^N \pi_k(a) p_k^*(a).$$

Economic meaning

A state contributes more to a long-run average when the process spends more time there. Both the state outcomes and the time shares may respond to AI.

Why more AI lowers long-run skill

The proof is an ordered comparison of adjacent stationary odds.

More AI weakly lowers effort at every skill state

For $a_\ell < a_h$,

$$e_i(a_\ell) = (x^\star - s_i - a_\ell)_+ \geq (x^\star - s_i - a_h)_+ = e_i(a_h).$$

Since λ is increasing,

$$\lambda(e_i(a_\ell)) \geq \lambda(e_i(a_h)) \quad \text{for every } i.$$

Economic meaning

Lower AI preserves at least as much human effort at every skill level, so every upward learning clock is at least as fast.

Every adjacent stationary odds ratio moves against high skill

Let

$$r_i(a) := \frac{\pi_{i+1}(a)}{\pi_i(a)} = \frac{\lambda(e_i(a))}{\mu}.$$

Then $a_\ell < a_h$ implies

$$r_i(a_\ell) \geq r_i(a_h) \quad \text{for every edge } i.$$

Economic meaning

Moving from low AI to high AI makes every step up the skill ladder less favorable relative to the step down.

First-order stochastic dominance formalizes “more high skill”

The low-AI distribution dominates the high-AI distribution when

$$\sum_{i=1}^k \pi_i(a_\ell) \leq \sum_{i=1}^k \pi_i(a_h) \quad \text{for every } k.$$

Economic meaning

For every cutoff s_k , low AI gives a smaller probability of being at or below that cutoff. Equivalently, it places more mass on high skill.

Read the inequality in the correct direction

The states are ordered from low to high. A smaller lower-tail probability means a better skill distribution.

A three-state example shows the downward shift

Suppose the adjacent ratios are

$$(r_1^\ell, r_2^\ell) = (2, 1.5), \quad (r_1^h, r_2^h) = (1, 0.75).$$

The unnormalized weights are therefore

$$w^\ell = (1, 2, 3), \quad w^h = (1, 1, 0.75).$$

After normalization,

$$\pi^\ell = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2} \right), \quad \pi^h = \left(\frac{4}{11}, \frac{4}{11}, \frac{3}{11} \right).$$

Economic meaning

High AI roughly doubles the probability of the lowest state and nearly halves the probability of the highest state.

Deskilling is certain; lower productivity is not

The skill result is unambiguous:

$a \uparrow \implies$ stationary skill distribution shifts downward.

But productivity contains two forces:

$\underbrace{p_k^*(a) \uparrow}_{\text{direct gain}}$ and $\underbrace{\pi(a) \text{ shifts down.}}_{\text{composition loss}}$

Bottom line

Skill degradation is the mechanism that makes a paradox possible, not a sufficient condition for the paradox by itself.

When does deskilling dominate?

Activity regimes isolate a smooth comparison between the direct and learning channels.

Active effort states always form a lower block

Effort is positive exactly when

$$e_k(a) > 0 \iff s_k + a < x^*.$$

Because $s_1 \leq \dots \leq s_N$, if state s_k is active, then every lower state is active too.

Active states = $\{1, \dots, m\}$ for some m .

Economic meaning

High-skill agents stop exerting effort first because their baseline input reaches the target x^* sooner.

An activity interval keeps the same states active

Define

$$I_m = ((x^* - s_{m+1})^+, (x^* - s_m)^+].$$

For a inside I_m ,

$$e_k(a) > 0 \ (k \leq m), \quad e_k(a) = 0 \ (k \geq m + 1).$$

Economic meaning

Inside one interval, no state switches formula. This lets us compare the two economic forces without being distracted by kinks at effort zero.

Active and inactive states react differently to AI

Inside I_m ,

$$p_k^*(a) = \begin{cases} p(x^*), & k \leq m, \\ p(s_k + a), & k \geq m + 1. \end{cases}$$

Active states

AI replaces effort. Output stays fixed, but learning slows.

Inactive states

Effort is already zero. AI directly raises output.

Bottom line

The paradox is created by the coexistence of replacement states and expansion states.

The aggregate derivative has a gain and a composition term

Differentiate the stationary average:

$$\mathcal{P}'(a) = \underbrace{\sum_{k=1}^N \pi_k(a) \frac{dp_k^*(a)}{da}}_{\text{within-state direct gain} \geq 0} + \underbrace{\sum_{k=1}^N \pi'_k(a) p_k^*(a)}_{\text{composition effect}}.$$

Because $\sum_k \pi'_k(a) = 0$, the second term only reallocates mass between states.

Economic meaning

The direct effect asks what changes within each skill group. The composition effect asks how the population moves across groups.

The sensitivity gap compares proportional responses

At the right endpoint of regime I_m , define

$$\Delta_m = \underbrace{\sum_{j=1}^m \frac{\lambda'(s_m - s_j)}{\lambda(s_m - s_j)}}_{\text{learning-weight sensitivity}} - \underbrace{\frac{p'(x^* + s_{m+1} - s_m)}{p(x^* + s_{m+1} - s_m) - p(x^*)}}_{\text{productivity-premium sensitivity}}.$$

Economic meaning

Δ_m is not a level comparison. It compares how rapidly the learning channel and the direct productivity channel respond locally to one more unit of AI.

The first term asks how fragile learning weights are

For a positive function f ,

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} \log f(x).$$

Therefore,

$$\sum_{j=1}^m \frac{\lambda'(s_m - s_j)}{\lambda(s_m - s_j)}$$

is the combined proportional sensitivity of the active learning rates.

Economic meaning

If this term is large, a small reduction in effort causes a large percentage reduction in the stationary weights supporting higher skill.

The second term asks how responsive the AI premium is

Let

$$R_m = p(x^* + s_{m+1} - s_m) - p(x^*) > 0.$$

Then

$$\frac{p'(x^* + s_{m+1} - s_m)}{R_m}$$

is the marginal direct gain relative to the existing productivity premium.

Economic meaning

If this term is large, additional AI expands output rapidly in the first inactive state. The direct productivity channel is then difficult for deskilling to overturn.

A positive gap means learning is locally more sensitive

$$\Delta_m > 0 \iff \text{learning sensitivity} > \text{productivity sensitivity.}$$

$$\Delta_m > 0$$

The negative learning channel can dominate when skill decay is sufficiently fast.

$$\Delta_m < 0$$

The direct productivity channel is locally more sensitive, so productivity remains increasing in the theorem's regime.

Bottom line

The sign identifies which channel is more responsive; μ determines how strongly that responsiveness enters stationary outcomes.

The decay rate controls the economic importance of learning

The key relative rate is

$$\frac{\lambda(e_k(a))}{\mu}.$$

Small μ

Skill is persistent. The process spends most time near high skill, so modest learning changes have limited aggregate impact.

Large μ

Skill depreciates rapidly. Maintaining high skill requires repeated learning, so reduced effort becomes consequential.

Bottom line

$\Delta_m > 0$ creates the vulnerable margin; sufficiently large μ makes that margin visible in long-run productivity.

Within one regime, productivity can turn only once

The paper proves that on every nonempty I_m ,

$\mathcal{P}(a)$ is either increasing everywhere, or increasing and then decreasing.

Economic meaning

As AI rises within a regime, the marginal direct gain becomes weaker while the composition loss becomes relatively stronger. Once the derivative turns negative, it cannot turn positive again before the next activity boundary.

Why regime boundaries matter

At a boundary, one state stops exerting effort and changes from the replacement formula to the expansion formula. The derivative may therefore reset.

The theorem converts intuition into a shape result

For sufficiently large μ and a nonempty activity interval I_m :

Paper's Theorem 1

1. If $1 \leq m \leq N - 1$ and $\Delta_m > 0$, then $\mathcal{P}$ increases and then decreases on I_m .
2. If $m \in \{0, N\}$ or $\Delta_m < 0$, then $\mathcal{P}$ is weakly increasing on I_m .

Bottom line

The paradox requires an intermediate regime, a positive sensitivity gap, and sufficiently rapid skill decay.

The two boundary regimes cannot create the paradox

I_N : all states active

Every state fills the gap to x^* :

$$p_k^*(a) = p(x^*).$$

Hence $\mathcal{P}(a) = p(x^*)$ is flat.

I_0 : all states inactive

Every effort is zero, so learning rates are fixed at $\lambda(0)$:

$$\mathcal{P}'(a) = \sum_k \pi_k p'(s_k + a) \geq 0.$$

Economic meaning

A decline needs some states to lose learning effort while other states receive a direct AI productivity gain.

Two states reveal the entire paradox

One learning rate, one decay rate, and one changing composition weight are enough.

In the intermediate regime, one state learns and one expands

For $N = 2$ and $a \in I_1 = (x^* - s_2, x^* - s_1]$,

$$e_1(a) = x^* - s_1 - a, \quad e_2(a) = 0.$$

Define $L(a) = \lambda(e_1(a))$. Then

$$\pi_1(a) = \frac{\mu}{\mu + L(a)}, \quad \pi_2(a) = \frac{L(a)}{\mu + L(a)}.$$

Economic meaning

Higher AI lowers $L(a)$, so stationary mass moves from the productive high-skill state s_2 toward the low-skill state s_1 .

Two-state productivity is a transparent weighted average

State 1 remains at the target output $P_0 = p(x^*)$, while state 2 produces $P_2(a) = p(s_2 + a)$. Thus

$$\mathcal{P}(a) = \frac{\mu P_0 + L(a)P_2(a)}{\mu + L(a)}.$$

Economic meaning

AI raises $P_2(a)$ directly, but it simultaneously lowers the weight $L(a)/(\mu + L(a))$ placed on that high-productivity state.

Bottom line

The whole paradox is visible in one quotient: the payoff rises while its weight falls.

The exact two-state threshold separates the two forces

Let

$$d = s_2 - s_1, \quad R = p(x^* + d) - p(x^*), \quad p'_R = p'(x^* + d).$$

When $\Delta_1 > 0$, define

$$\bar{\mu}_2 = \frac{\lambda(0)^2 p'_R}{\lambda'(0)R - \lambda(0)p'_R}.$$

Economic meaning

The denominator is positive exactly when learning is more sensitive than the direct productivity premium. The numerator scales the remaining direct gain.

Two conditions produce the N-shaped response

For two states:

$$\Delta_1 > 0 \quad \text{and} \quad \mu > \bar{\mu}_2$$

imply the global sequence

flat $\longrightarrow$ rising $\longrightarrow$ falling $\longrightarrow$ rising.

Economic meaning

At low AI, effort absorbs assistance and output is flat. In the intermediate regime, direct gains initially win and deskilling later wins. At high AI, effort is already zero, so only direct gains remain.

A numerical calibration gives a concrete threshold

Use

$$\lambda(e) = 1 + e, \quad p(x) = \log(1 + x), \quad x^* = 3, \quad (s_1, s_2) = (0, 2).$$

Then

$$R = \log 6 - \log 4 \approx 0.4055, \quad p'_R = \frac{1}{6},$$

and

$$\bar{\mu}_2 = \frac{1/6}{0.4055 - 1/6} \approx 0.698.$$

Economic meaning

With $\mu = 1$, a declining subregion exists. With $\mu = 0.5$, memory is persistent enough that productivity remains increasing.

The productivity shortfall can be arbitrarily severe

The paper proves that for every $\varepsilon > 0$, one can choose valid primitives such that

$$\inf_{a_\ell < a_h} \frac{\mathcal{P}(a_h)}{\mathcal{P}(a_\ell)} \leq \varepsilon.$$

Economic meaning

This is an existence result. It says the model does not impose a universal lower bound on the damage from deskilling; it does not predict a severe decline for every empirical setting.

Construction intuition

Make zero-effort learning almost impossible, but let a small amount of effort be enough to sustain high-skill stationary mass.

Simpson's paradox is the right aggregation analogy

Differentiate the aggregate:

$$\mathcal{P}'(a) = \underbrace{\sum_k \pi_k(a) p_k^{*'}(a)}_{\text{within-state effect} \geq 0} + \underbrace{\sum_k \pi'_k(a) p_k^*(a)}_{\text{composition effect}}.$$

Economic meaning

Every skill group can improve while the aggregate falls, because AI changes the shares of the groups. The reversal comes from endogenous weights.

Bottom line

The model's paradox is Simpson-type: the direction within groups differs from the direction after aggregation.

A numerical Simpson reversal needs almost no algebra

Under low AI, suppose

$$\pi^\ell = (0.25, 0.75), \quad p^\ell = (10, 16), \quad \mathcal{P}_\ell = 14.5.$$

Under high AI, both state productivities rise by one, but the weights reverse:

$$\pi^h = (0.75, 0.25), \quad p^h = (11, 17), \quad \boxed{\mathcal{P}_h = 12.5 < 14.5.}$$

Economic meaning

Each group gains one unit, yet the population spends much more time in the low-productivity group. The composition loss is three units, which dominates the direct gain of one.

Exogenous transition rates remove the paradox

If upward rates are fixed numbers λ_k rather than functions of effort, then

$$\pi'_k(a) = 0.$$

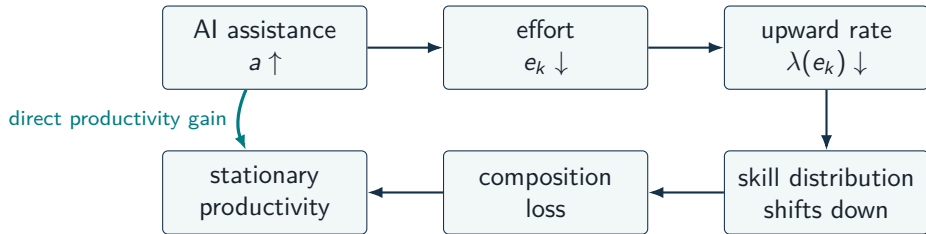
Therefore,

$$\mathcal{P}'(a) = \sum_{k=1}^N \pi_k \frac{dp_k^*(a)}{da} \geq 0.$$

Economic meaning

Heterogeneous skill alone is not enough. The paradox requires AI to change effort, effort to change learning, and learning to change the long-run skill distribution.

The complete causal chain has two competing paths



How to read the theorem

Δ_m compares the local sensitivities of the two paths. The decay rate μ determines how strongly the learning path affects long-run weights.

Six conclusions to carry out of Part II

1. AI and effort are perfect substitutes in the static problem.
2. Effort is an input into skill development through $\lambda(e)$.
3. More AI lowers every upward stationary odds ratio and shifts skill downward.
4. Aggregate productivity combines a positive direct effect and a potentially negative composition effect.
5. A positive sensitivity gap and sufficiently rapid decay create a declining region.
6. The paradox disappears when transition rates are exogenous.

Bottom line

The central message is not “AI lowers productivity.” It is that static gains do not determine long-run outcomes when the weights are endogenous.

Questions that test economic understanding

1. Why is $\lambda(e)$ a rate rather than a probability?
2. Why does $\pi_{k+1}/\pi_k = \lambda(e_k)/\mu$ summarize one skill edge?
3. Why does lower AI first-order stochastically dominate higher AI?
4. Why is deskilling not sufficient by itself for a productivity decline?
5. What distinct roles do Δ_m and μ play?
6. Why can the derivative become positive again after an activity boundary?
7. What assumption does the exogenous-transition benchmark remove?

A strong answer should describe the economic mechanism before manipulating the equation.

References and scope

Primary paper

Ali Aouad, Thodoris Lykouris, and Huiying Zhong. *Human–AI Productivity Paradoxes: Modeling the Interplay of Skill, Effort, and AI Assistance*. arXiv:2605.11350v1, 2026.

Probability background

Geoffrey Grimmett and David Stirzaker. *Probability and Random Processes*, 3rd ed. Oxford University Press, 2001.

Deck scope

This lecture covers endogenous skill development and the first productivity paradox. It emphasizes interpretation rather than reproducing every algebraic step in the longer tutorial.

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